In our response to reviewers below we highlight reviewer comments in green, and our responses in black. New text added to the manuscript is highlighted. We have formatted a page break between each reviewer. Where necessary, we have interjected our responses among reviewer comments.

Referee 1:

Summary:
This study describes a novel phylogenomic dataset (anchored hybrid enrichment) for microhylid frogs (Anura: Microhylidae), a family with several unresolved higher-level relationships. It has a special focus on Asterophryinae and understanding the origins of the Australian members of this subfamily. Overall, we thought the manuscript was well-written and concise. Strengths of the manuscript include the figures (which are exceptional), a clear introduction into microhylid systematics, and a novel and interesting phylogenomic dataset. While we found the paper to be noteworthy and publishable, we also identified several areas for improvement which we describe below.

Thank you for taking the time to review our manuscript, and for sharing the process with a student. It's the best way to learn. We appreciate your input and support of our manuscript and endeavour to address all your identified areas of improvement.

General comments:
Morphological analyses: We identified two main issues with the morphological data and analyses as currently presented. First, as described in the text, the data are taken from previous studies, yet the authors did not address data compatibility across these datasets. Interobserver variation in character measurement is a known issue in frog morphometrics (see Hayek et al. 2001, Alytes, Frog morphometrics: a cautionary tale).

This is a really interesting point and also one that is extremely difficult to address. The Haytek et al. (2001) paper highlights that morphological data can be highly sensitive to both inter- and intraobserver bias. As a result, measurement error can go on to affect downstream interpretations. As our dataset comprises measurements taken across multiple decades by up to 13 different observers, there is little ability to measure consistency, accuracy, and precision, across all of these reported data. Unfortunately, getting a single observer to measure dozens of individuals 20 or more times for hundreds of species is unrealistic. In Haytek et al. (2001), the focus of the paper is sensitivity in intraspecific sampling. While we include intraspecific sampling in our dataset (none of which were originally published with supporting measurement error), our focus is above the species level. In this case the expectation is that interspecific variation (including error) is greater than intraspecific variation (including error), and that any error is random. To test this idea, we randomly downsampled our morphological dataset to 11 species with more than five individuals measured per species (total 94 individuals). We used this dataset to run a separate Random Forest analysis where the classifier was the species. Of these 11 species, only one (Cophixalus balbus) had a classification error > 0. Looking closer, this is driven by a single individual which is not confidently assigned to the correct species (<0.70). Given that of 94 individuals across 11 species, only a single individual is misidentified, we can feel relatively confident that our assumption that interspecific (and inter-generic) variation is greater than intraspecific variation is true. We believe this provides evidence of the reliability of the data for our purpose. For full transparency, we include this information in the Supplement.

It also is not clear exactly how the measurements were taken. For example, what landmarks were used to define head width (this differs across some anuran studies)? A table with summary statistics for the measurements along with references would be helpful and improve interpretability of the morphological results. This could be a supplementary table. Second, many of the diagnostic features that have been used to develop the taxonomy of Asterophryinae are based on soft or internal anatomy (e.g. vertebral column morphology or tongue morphology). These characters are not considered as part of the morphometric analysis presented here, yet the authors seem to conclude that morphology is (in general) quite bad at delimiting diversity based on their analysis. Many of the eight morphological measurements they use are highly conserved across anurans or associated with ecological habitats and not taxonomy (e.g. eye diameter; Thomas et al. 2020, Proc. Roy. Soc. B). We encourage the authors to either revise their text or include a broader comparative examination of morphology.

Despite the species richness of Asterophryinae, the taxonomy of the group and morphological assessments have primarily been carried out by a small number of researchers who often work collaboratively with one another. These researchers (Günther, Kraus, Allison, Oliver, Richards, Zweifel) account for the overwhelming majority of measurements in our morphological dataset. As a rule, they collect the same series of traits, measured in the same way. These follow the measurements of Zweifel, as described in Zweifel (1972, 2000). To emphasise this, we include how these traits are measured on microhylid frogs in the Supplement. The raw data, including the source publication for each sample and its measurements are included in the GitHub repository found here: github.com/IanGBrennan/Asterophryinae/blob/main/Data/Asterophryinae_RAWdata.csv (this was in the original submission).

While we agree that some traits are often quite conserved and others are associated with ecology, they are the traits commonly measured in Asterophryinae taxonomic papers, where the data was collected from. Assessment of these traits is relatively quick and non-destructive, contra to clearing/staining, skeletal preparation, or investigation of musculature. As a result, these are the characters most often used in a first assessment of a new individual or species. We want to stress that we don't think "morphology is bad at delimiting diversity". Morphology is an integral tool for understanding biodiversity. Unfortunately, this set of commonly-used morphological data is not always reliable for delimiting genera, at least within Asterophryinae frogs.

We agree that internal anatomical traits are often used to delimit genera or species. It is unclear, however, what is causing the breakdown of Asterophryinae taxonomy through poly and paraphyletic genera. If all new species were being thoroughly examined both internally and externally before being assigned to a genus, then we might make the judgement that existing genera diagnoses are non-exclusive and poorly defined. However, if some or any new species are not examined entirely, and instead assigned to a genus based on general similarity to existing congeners, then the cause of the paraphyly is taxonomic misassignment.

Taxonomic confusion in Asterophryinae is likely a result of both processes working in concert, compounded by repeated taxonomic updates that have further muddled the situation. We comment on this further in the updated Discussion. Further, it is unclear if internal anatomical traits provide better signal than external linear measurements. Generating the data for a better comparison is unfortunately outside the scope of this study.

Here is one of the relevant (and currently uncited) studies that may help with a deeper examination of asterophryine morphology:Burton, T. C. 1986: A reassessment of the Papuan subfamily Asterophryinae (Anura: Microhylidae). Records of the South Australian Museum 19:405–450.

Thank you for bringing this work to our attention, it's a really impressive paper that we are embarrassed to have missed. While it is clear this research required a tremendous amount of work and generated a lot of great data, it is difficult to assess what use it might be to us. The character states in Table 4 would be fantastic data to include in our work, but unfortunately character states are summarized to genus instead of species. For example, the genus Phrynomantis as described in Burton (1986) now represents species in four separate genera (Asterophrys, Callulops, Hylophorbus, and Mantophryne).

Missing citation: Portik et al. (2023, Mol. Phylogenet. Evol.) also discuss microhylid phylogenomics and issues within Asterophryinae. This study features several similar results that seem relevant for the discussion provided in this manuscript (e.g. placement of Aphantophryne pansa with Cophixalus and a well-supported Microhylinae + Dyscophinae).

Thank you very much for pointing out this paper, which we should not have missed. It also jogged our memory that there is a second Portik et al. (2023) paper in Molecular Biology and Evolution that we should've also cited. We have now included these papers.

Figure 1 data: The species richness and geographic patterns of diversity are not described in the text. Where these data come from and how they were analysed should be discussed in the text.

The details of our species richness maps were an unfortunate omission. We have added a new section to the Materials/Methods section titled Visualising Species Richness which briefly outlines our strategy and directs readers to the supplement.

> To summarise geographic patterns of species richness in microhylid frogs we started by downloading species range maps for all Anurans from the IUCN website. We then subset these data separately to the family Microhylidae, Asterophryinae genera, and the genera Austrochaperina and Cophixalus. We summed richness at a resolution of one quarter degree and visualised the resulting raster using ggplot2 and ggmap2. All code is available in the supplement.

Data availability: Are the phylogenomic and morphological data available? There is no mention of GenBank SRA or where the morphology matrix may be obtained.

All data is available in the project GitHub repository at https://github.com/IanGBrennan/Asterophryinae . This information is included in the text on line 163 (previously Line 158): “Sample, alignment, and gene trees are available alongside all other materials Github …”

Specific comments by line number:
Line 57: Confusing use of ‘ranging’ . We think it would be clearer if the others used ‘occur’.

We have changed the wording to read “occuring from North …”

Line 62: I am not sure that I would agree that the consensus is for most subfamilies (i.e. intrafamilial relationships). The cited studies vary in their placement of several subfamilies and for some subfamilies (e.g. Melanobatrachinae) there are no phylogenomic data.

We have relaxed the language here to emphasise that there is consensus for some relationships, but not all. The passage now reads:

> The increasing size of molecular datasets has found support for recognized subfamilies and provided evidence for some intrafamilial relationships, despite using different marker types (Feng et al. 2017; Streicher et al. 2020; Hime et al. 2021).

Line 80-81: Confusing sentence, should be something like, colonization event…’ ‘ …represent a single

We have adjusted this sentence to be clearer about the competing theories surrounding Australian microhylids.

> The evolutionary history of microhylid frogs in Australia is currently unresolved, including whether mainland Australian species are the result of repeated dispersals from New Guinea, or instead represent Australian clades.

Line 174: What 40 publications exactly? These should be listed either in text or as an Appendix.

The 40 publications from which we collected data are listed in the morphological data file under the header Source. We now refer the reader to the supplement for this list. These data are available at:
https://github.com/IanGBrennan/Asterophryinae/blob/main/Data/Asterophryinae_RAWdata.csv

Line 180: Additional random forest analysis details are required. For example, the number of trees.

We have amended this sentence to provide more detail:

> We then used current generic assignments as a discrete character and carried out a RandomForest (Liaw & Wiener, 2022) analysis across 10,000 decision trees on the morphological traits to determine rates of mischaracterization based on gross phenotype (genus ~ morphology)

Line 184: What criteria were used to determine that only the first two axes should be examined in plots for the PCA, LDA, and FDA?

The first two PC axes were used exclusively for visualisation and not for data analysis. Together they account for 89% of the variance in the morphological data (see Fig.3), which we deemed appropriate for representing the data. We provide more detail in an updated version of this sentence:

> To visualize the partitioning of morphological space we used dimensionality reduction techniques (PCA, LDA, FDA) and plotted the first two axes of variation, which together account for ~89%.

Line 205: ‘Tree topology’ is redundant. Just use one.

Now reads “Our topology …”

Line 218: If Mantophryne was not sampled, how can you comment on its placement? From the mitochondrial analysis with the Hill et al. (2022) data?

Correct. We realise this phrasing is confusing. We have added a sentence to clarify:

> While not included in this work, there is consistent support from other datasets that Mantophryne also sits in this clade (Morris et al. 2026).

We believe this better explains that while not sampled here, there is evidence that Mantophryne belongs to this group.

Line 234: How many species were represented in each of the genera that had their morphology compared?

All morphological data was made available through the supplement in the GitHub repository. A summary of the number of species per genus sampled is provided below: Aphantophryne (3), Asterophrys (4), Austrochaperina (16), Barygenys (3), Callulops (16), Choerophryne (7), Cophixalus (43), Copiula (7), Gastrophrynoides (2), Hylophorbus (10), Mantophryne (2), Oreophryne (9), Paedophryne (4), Siamophryne (1), Sphenophryne (5), Vietnamophryne (3), Xenorhina (8). We now provide this information in the Supplement in the section Morphological Sampling.

Line 239 / Figure S4: When genera were misclassified, which incorrect groups did the RandomForest method, LDA, and FDA place them with?

Due to the number of comparisons (17 genera X 17 genera), this is difficult to summarise neatly here. To share this information, which may be of interest, we have generated a pairwise model classification table and included that in the Supplement. Misclassifications are typically quite consistent across the three methods, e.g. RF, LDA, and MDA all misclassify five Cophixalus samples as Oreophryne.

Line 264: How can you differentiate between elevated Cretaceous extinction versus low lineage birth during the Cretaceous?

This is a great question. Without a reliable fossil record, it is certainly difficult to distinguish between these two processes. We specifically used the language “likely indicative”, which we believe conveys some uncertainty in our contention. In our opinion, it is more likely that a long branch leading to the radiation of a group is a sign of extinction in the group’s history and not the maintenance---without speciation---of a single taxon for tens of millions of years. We base this contentionon the organismal fossil record and Raup's ideas on the ubiquity of extinction in animal evolutionary history. We also find support for this idea in the elevated extinction of many other groups at the Cretaceous-Paleogene boundary, one of Earth's most extreme mass-extinction events.

Line 317: Missing multiple key morphological studies at the genus level (e.g. Burton study mentioned previously).

Thanks for catching this, we have included a set of the relevant studies now:

> (Zweifel 1972; Zweifel 1985; Burton 1986; Zweifel 2000)

Line 335: There are no Mantophryne sampled in the main phylogenomic study, so please be clear about what is meant here.

Yes. We have added text that highlights that Mantophryne was not sampled here, but likely belongs to this clade.

Line 408: Cophixalus is labelled as "top" and Austrochaperina is labelled as "bottom," but they are both on the bottom, located left and right of each other respectively. Additionally, in all other instances the location precedes what is described while this line has the location afterwards. For consistency, something could be written like, "… species richness in (right) Cophixalus and…"

We have corrected this to reflect that Cophixalus is plotted on the left and Austrochaperina is plotted on the right.

Line 484: Note tab formatting issue with references (Hill and Hime references).

Thanks for catching this.

Figure 1: Where do these species richness data come from? Not addressed at all in the Text.

Please see our response to a similar comment above. We have included text to the Materials and Methods section of the manuscript that describes where we collected the data from, and how we generated these species richness maps.

Figure 3: Datapoints appear 'ghostly' on some computer screens. The main data points should be darker so that the entire dataset is visible to a wide range of readers. Also, some of the frog silhouettes are covering data points and obscure the information being presented.

Thank you for pointing this out. To clean up this figure and make it more easily readable, we have removed the transparency filter on the background points that was making them appear "ghostly" and outlined them to distinguish between multiple points in a small area. We have also moved the frog silhouettes so no points are being obscured.

Referee 2:

It was a pleasure to read this paper that is focused on microhylid frog phylogeny. The quality of the illustrations (including Supplementary Figure 2) is outstanding and complements the solid scientific results very well. I was especially intrigued to see they incorporated statistical analyses of morphological characters of the genera to show that they can be unreliable to assign a given species to genera. The only minor flaw I found with the study, and it is not a fatal one, is that the authors do not cite the important study of Portik et al. (2023) on frog phylogeny: https://doi.org/10.1016/j.ympev.2023.107907. Results of this latter study have important comparisons to the current one, including other similar comparisons around line 200. Note this passage from the Discussion: "Among asterophryines (Fig. 12), four genera were non- monophyletic: Austrochaperina, Cophixalus, Copiula, and Oreophryne. Hill et al. (2022) also found these genera to be non-monophyletic. Furthermore, our tree showed Oninia as nested within Asterophrys. These results show that the genus-level taxonomy of asterophryines needs revision." It should at least be mentioned in the Intro.

We appreciate the positivity around our manuscript. Thanks for pointing out that we missed that Portik et al. (2023) paper (also pointed out by Referee 1). The passage you highlight is certainly relevant and we have sought to address this in the discussion of our main text where we reiterate the need for careful genus-level taxonomic revision.

The Discussion is well-written and informative, but there were three areas that I thought could be elaborated a bit more to make the paper of broader interest. The incredible diversification of this group near the Cretaceous-Paleogene boundary is a common theme for other nocturnal vertebrates, including geckos, bats, and other groups of frogs. Could something about this be added to the Discussion? Although oceanic rafting events are known in geckos, iguanian lizards, and other reptile groups, I wonder how rare it is in frogs and how limited the distances might be. Work by Miguel Vences, Breda Zimkus and Rayna Bell provide examples around Africa, including Madagascar. There is also work by several people regarding dispersal of frogs among Asian islands. Last but not least, climate is mentioned only once on line 378. Given the timing of diversification of this group in the Miocene, I wonder if global cooling played a role in changing habitats and creating refugia? Of course if any of this is beyond the scope of the current paper, the authors are free to ignore my suggestions here.

It's always great to see what the target audience of our research thinks should be included. We agree these are interesting topics and so have attempted to address them further in the text. We have incorporated elements that speak to KPg extinction:

> In light of this, the long stem branch leading to the Microhylidae (>35 ma) is likely indicative of elevated Cretaceous extinction and a dramatic rebound in the Paleogene. This pattern is seen in many vertebrate groups, such as squamate reptiles (Longrich et al. 2012), mammals (Meredith et al. 2011), and frogs (Feng et al. 2017), and potentially highlights biased survival of nocturnal groups (Wu et al. 2017).

And overwater dispersal:

> As a result, microhylids either undertook two separate dispersals from Madagascar to Asia and Australia, or a single dispersal followed by a return of the Dyscophinae to

> Madagascar, a trip of ~4,000 km. Long distance dispersals are not unheard of in amphibians (Fonte et al. 2019), but these distances over open ocean would be exceptional.

We would love to explore the role a changing global climate may have played in the diversification of this group, but it feels that with our limited sampling it might be premature. The Mid Miocene Climatic Optimum has been implicated in driving the diversification of many groups and lines up roughly with the earliest Asterophryinae divergences. This connection is limited/speculative and more thorough modelling should be carried out before claiming links between the two. Perhaps we will be better placed to test these ideas in the future.

Aside from this, my comments below are quite minor. I hope they are helpful to the authors and I look forward to seeing this paper in print. I do not wish to remain anonymous: Eli Greenbaum

Thank you Eli, we appreciate your considered and very constructive comments.

Line 54 and elsewhere in the MS: update to Frost 2026

We have updated this.

Line 74: if two new species are going to be mentioned they should probably be listed- I assume these are Callulops gobakula Hoskin, 2025 and Choerophryne koeypad Hoskin 2025.

Yes, that is correct. We have kept this information in the introduction and added the species.

Line 131: ensure the past tense is consistent in this paragraph

Thank you. We have amended this awkward passage to be consistent in tense.

Line 218: and should not be in italics and there seems to be a grammatical error at the end of the sentence that makes its meaning confusing

We have corrected this.

Line 222: well-defined here, and in other areas of the MS, well-supported

Changed to “well-supported”.

Line 239: Fig.3 should be Fig. 3.

Corrected to “Fig. 3”.

Line 246: third-largest

Corrected to “third-largest”.
Line 340: which study identified the Oreophryne 'B' clade?

The two sources cited earlier in the sentence, (Rivera et al. 2017; Hill et al. 2022).

Line 344: this should be Dubois 2021

Corrected to “Dubois et al. (2021)”.

Line 421: best to not start a sentence with a number

Fair call. We have corrected this to “The 95%”.

Figure 1: the authors might consider adding scale bars to the maps so that relative distances can be assessed.

We have added scale bars the maps as requested.

Figure 2: I see Sphenophryne thomsoni is listed with a slash as "Geny", I assume, Genyophryne. I realize the taxonomic placement of this species is problematic when I look at the tree, but could the authors add a brief explanation to the caption to avoid confusion about this?

To keep consistent with other nomenclatural changes, we have dropped the “Geny” label associated with this taxon and refer to it solely as *Sphenophryne thomsoni*.

Figure 3: Latin names should be in italics

We have corrected this.